

Assignment #5: Initialization

Atmospheric Sciences 528: Atmospheric Data Analysis

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100 pts

Instructions

1. Perform a single pass Barnes analysis of the geopotential heights in the data set given to you earlier. I have modified the stations to remove those that do not report winds, so use all remaining 132 stations in the data set. The parameters you used in your previous assignments are the same here:

- a. Polar Stereographic Projection Parameters

$$\varphi_0 = 60^\circ N$$

$$\lambda_0 = 100^\circ W$$

$$m = 1 / 15000000$$

- b. Analysis Grid Parameters

$$x_0 = 25 \text{ cm}$$

$$y_0 = -15.5 \text{ cm}$$

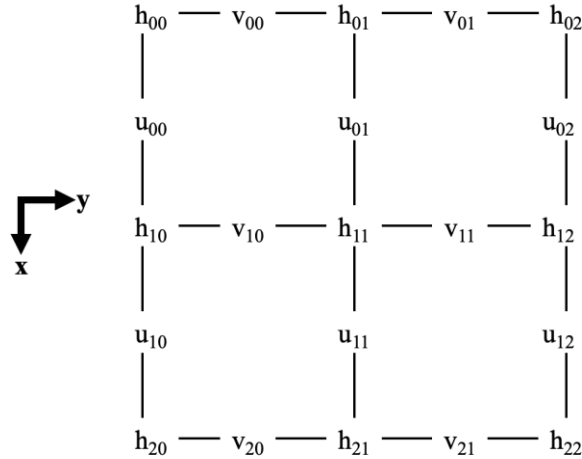
$$\Delta x = \Delta y = 1.27 \text{ cm}$$

The one-pass Barnes analysis of the observation values is what you did for the statistical objective analysis background with the weights determined using the above weight function:

$$w(d_{ik}) = \exp\left(\frac{-d_{ik}^2}{K_d}\right)$$

where $K_d = 10.8844524 \text{ cm}^2$. Use $r_e = 12.7775892 \text{ cm}$ as your radius of influence.

2. Next, set up an staggered grid where the u and v wind components are staggered from the height field using the Arakawa C-grid below. The spacing between the u- and v- grid points are still $\Delta x = \Delta y = 1.27 \text{ cm}$, with the u-grid points keeping the same y-grid points as the height grid points but exhibiting x-grid point that are offset from the height grid by $\frac{1}{2} \Delta x$. Similarly, the v-grid points will have the same x-grid points as the height grids, but y-grid points that are offset from the height grid by $\frac{1}{2} \Delta y$. Note, to achieve this, the u-grid will have dimensions (nx-1, ny) and the v-grid will have dimensions (nx, ny-1).



3. Convert the observed winds to the polar stereographic projection using the following equations from the Polar Stereographic Projection document. First, converting from wind speed (C) and direction (α):

$$U = -C \sin \alpha ,$$

$$V = -C \cos \alpha ,$$

Then, rotating the winds to the new projection using the latitude deviation ($\lambda = \lambda_{stn} - \lambda_0$):

$$u = -U \sin \lambda - V \cos \lambda ,$$

$$v = U \cos \lambda - V \sin \lambda .$$

4. Analyze the u- and v-winds on the staggered u- and v-grids, respectively. Do this using the single pass Barnes algorithm from above.
5. Calculate a static Coriolis parameter using φ_0 . Now run the code to calculate geostrophic winds. Create a 2-panel figure with:
 - a. The analyzed heights and analyzed winds.
 - b. The analyzed heights and geostrophic winds.
6. Use the code to run the shallow water model for 12 hours with a 60s timestep for both the analyzed winds and geostrophic winds (i.e., two separate runs). Create the following plots:
 - a. A 2-panel figure with:
 - i. “Forecasted” height and winds using the analyzed wind field at 12 hours.
 - ii. “Forecasted” height and winds using the geostrophic wind field at 12 hours.
 - b. A 2-panel figure with:
 - i. Difference between the initial and “Forecasted” height and winds using the analyzed wind field at 12 hours.

- ii. Difference between the initial and “Forecasted” height and winds using the analyzed wind field at 12 hours.
- c. A line plot showing the mean absolute height tendency (dh/dt) in meters per hour for each simulation. I.e., there should be two lines – one for simulation using analyzed winds and one using geostrophic winds.

Questions & Deliverables

- 7. Either link your github repository to blackboard, or upload a compressed directory with:
 - a. Your code
 - b. 4 plots (1 with both initial conditions, 1 with both forecasts, 1 with both forecast differences, and 1 line plot of height tendency).
 - c. Answers to the following questions:
 - i. The only difference in the initial conditions is the wind field. Discuss the differences in the forecast height/wind fields? What might this suggest about forecasts from models that are sensitive to unbalanced initial conditions?
 - ii. Contrast the height difference fields. Does one seem smoother than the other?
 - iii. Describe the height tendency plot. Hypothesize why there are differences. What does this mean about forecast stability? Note, the advective scheme used for height tendency will have a smoothing effect in time (acts as a low-pass filter).